AM Electricals

Control panel manufacturer, Kolkata — CPRI type-tested, ISO 9001:2008 certified, founder-led since 1991.

34	5,000	13	70 kA / 1s
Years in business	Sqft Kolkata factory	Panel families	MCC / PCC withstand

Company profile prepared for prospective clients and procurement review. Covers facility, certifications, full product range, industries served and track record.

ESTABLISHED 1991

Founder-led since 1991.

AM Electricals was founded in Kolkata by Ajit Debnath in 1991 to build control panels for Bengal's growing industrial base. 34 years later we still make what we sell, from our own 5,000 sqft factory on Kamardanga Road.

The founder

Ajit started AM Electricals as a one-man workshop in 1991. Today he still reviews every non-standard job, every DG-sync SLD, every PLC panel wiring drawing. When an escalation lands, it lands on his phone. Three decades of returning customers who don't need to explain their site twice.

Registered details

Founded **1991**	Founder / CEO **Ajit Debnath**
Legal status **Proprietorship Firm**	Business **Manufacturer, Supplier, Service Provider**
Sector **Control Panels and Boards**	Factory address **7S Kamardanga Road, Near Anandapalit, Kolkata 700046**
Phone **+91-7003707013**	Email **amelectricals@yahoo.co.in**

THE WORKSHOP

Made where it's sold.

5,000 sqft permanent factory in Kolkata with 10 to 20 electrical engineers, draughtsmen, panel wiremen and test technicians. Every panel is built, wired, tested and FAT-signed here before dispatch. CPRI type-tested designs. ISO 9001:2008 quality process. IS 8623 / IEC 61439-2 as standard.


Panel testing before dispatch


Colour-coded busbar wiring


AMF control panel build


Finished panel, door open

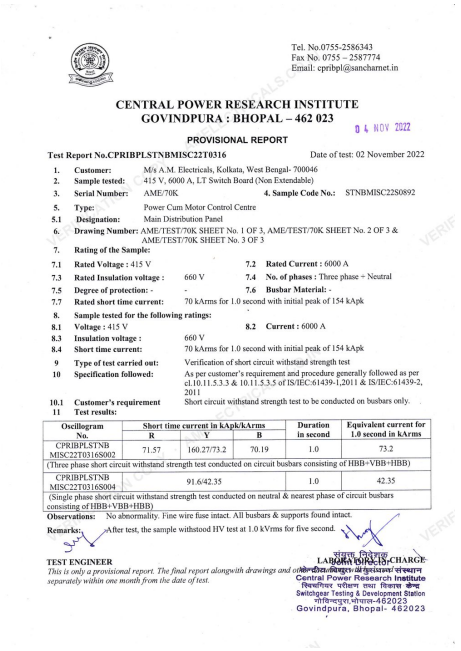

Contactor wiring detail


Fire pump panel, finished

PROOF, NOT CLAIMS

Certifications on file.

CPRI — Central Power Research Institute short-circuit withstand strength test, 70 kArms for 1.0 second with initial peak of 154 kApk (verification report, LT switch board, 6000 A, non-extendable). **ISO 9001:2008** — Quality Management System certified. **MSME** — Entrepreneurs Memorandum registration, filed 1991.



Tel. No.0755-2586343
Fax No. 0755 – 2587774
Email: cpribpl@sancharnet.in

CENTRAL POWER RESEARCH INSTITUTE
GOVINDPURA : BHOPAL – 462 023

04 NOV 2022

PROVISIONAL REPORT

Test Report No.CPRIBPLSTNBMISC22T0316 Date of test: 02 November 2022

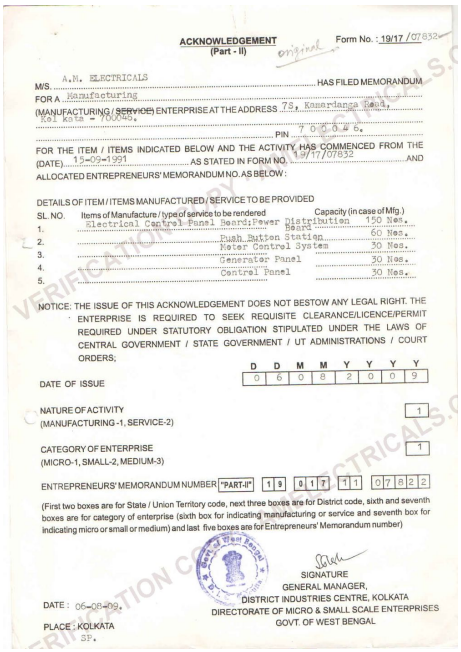
1. **Customer:** M/s A.M. Electricals, Kolkata, West Bengal- 700046
2. **Sample tested:** 415 V, 6000 A, LT Switch Board (Non Extendable)
3. **Serial Number:** AME/70K **4. Sample Code No.:** STNBMISC22S0892
5. **Type:** Power Cum Motor Control Centre
5.1 **Designation:** Main Distribution Panel
6. **Drawing Number:** AME/TEST/70K SHEET No. 1 OF 3, AME/TEST/70K SHEET No. 2 OF 3 & AME/TEST/70K SHEET No. 3 OF 3
7. **Rating of the Sample:**
7.1 **Rated Voltage :** 415 V **7.2 Rated Current :** 6000 A
7.3 **Rated Insulation voltage :** 660 V **7.4 No. of phases :** Three phase + Neutral
7.5 **Degree of protection:** - **7.6 Busbar Material:** -
7.7 **Rated short time current:** 70 kArms for 1.0 second with initial peak of 154 kApk
8. **Sample tested for the following ratings:**
8.1 **Voltage :** 415 V **8.2 Current :** 6000 A
8.3 **Insulation voltage :** 660 V
8.4 **Short time current:** 70 kArms for 1.0 second with initial peak of 154 kApk
9 **Type of test carried out:** Verification of short circuit withstand strength test
10 **Specification followed:** As per customer's requirement and procedure generally followed as per cl.10.11.5.3.3 & 10.11.5.3.5 of IS/IEC:61439-1,2011 & IS/IEC:61439-2, 2011
10.1 **Customer's requirement** Short circuit withstand strength test to be conducted on busbars only.
11 **Test results:**

Oscillogram No.	Short time current in kApk/kArms R	Y	B	Duration in second	Equivalent current for 1.0 second in kArms
CPRIBPLSTNB MISC22T0316S002	71.57	160.27/73.2	70.19	1.0	73.2
(Three phase short circuit withstand strength test conducted on circuit busbars consisting of HBB+VBB+HBB)					
CPRIBPLSTNB MISC22T0316S004	91.6/42.35			1.0	42.35
(Single phase short circuit withstand strength test conducted on neutral & nearest phase of circuit busbars consisting of HBB+VBB+HBB)					

Observations: No abnormality. Fine wire fuse intact. All busbars & supports found intact.
Remarks: After test, the sample withstood HV test at 1.0 kVrms for five second.

TEST ENGINEER **LABORATORY INCHARGE**
This is only a provisional report. The final report alongwith drawings and photographs will be issued separately within one month from the date of test.
Central Power Research Institute
Switchgear Testing & Development Station
Govindpura, Bhopal- 462023

CPRI test report



ACKNOWLEDGEMENT Form No. : 19/17 /07832
(Part - II)

M/S. A.N. ELECTRICALS HAS FILED MEMORANDUM
FOR A Manufacturing
(MANUFACTURING / SERVICE) ENTERPRISE AT THE ADDRESS 78, Kasaridanga Road,
Kol Kata - 700046,
PIN 7 0 0 0 4 6,
FOR THE ITEM / ITEMS INDICATED BELOW AND THE ACTIVITY HAS COMMENCED FROM THE (DATE) 15-09-1991 AS STATED IN FORM NO. 19/17/07832 AND
ALLOCATED ENTREPRENEURS' MEMORANDUM NO. AS BELOW :

DETAILS OF ITEM / ITEMS MANUFACTURED / SERVICE TO BE PROVIDED

SL. NO.	Items of Manufacture / type of service to be rendered	Capacity (in case of Mfg.)
1.	Electrical Control Panel Board(Power Distribution Board	150 Nos.
2.	Push Button Station	60 Nos.
3.	Water Control System	30 Nos.
4.	Generator Panel	30 Nos.
5.	Control Panel	30 Nos.

NOTICE: THE ISSUE OF THIS ACKNOWLEDGEMENT DOES NOT BESTOW ANY LEGAL RIGHT. THE ENTERPRISE IS REQUIRED TO SEEK REQUISITE CLEARANCE/LICENCE/PERMIT REQUIRED UNDER STATUTORY OBLIGATION STIPULATED UNDER THE LAWS OF CENTRAL GOVERNMENT / STATE GOVERNMENT / UT ADMINISTRATIONS / COURT ORDERS;

DATE OF ISSUE D D M M Y Y Y Y : 0 6 0 8 2 0 0 9

NATURE OF ACTIVITY (MANUFACTURING -1, SERVICE-2) 1

CATEGORY OF ENTERPRISE (MICRO-1, SMALL-2, MEDIUM-3) 1

ENTREPRENEURS' MEMORANDUM NUMBER "PART-II" 1 9 0 1 7 1 1 0 7 8 2 2

(First two boxes are for State / Union Territory code, next three boxes are for District code, sixth and seventh boxes are for category of enterprise (sixth box for indicating manufacturing or service and seventh box for indicating micro or small or medium) and last five boxes are for Entrepreneurs' Memorandum number)

SIGNATURE
GENERAL MANAGER,
DISTRICT INDUSTRIES CENTRE, KOLKATA
DIRECTORATE OF MICRO & SMALL SCALE ENTERPRISES
GOVT. OF WEST BENGAL

DATE : 06-08-09.
PLACE : KOLKATA

MSME registration

ISO 9001:2008 certificate and full CPRI test series available on request.

THE RANGE

13 panel families.

Every panel ships with a BOQ, SLD, GA drawing and FAT report. Built to spec for the site, not pulled off a shelf. Full detail for each family follows.

APFC Panels

Thyristor APFC, 25 to 800 kVAR, IEC 61439-2

Power factor below 0.95 draws a penalty on most industrial tariffs. An APFC panel switches capacitor banks in steps to hold the plant's power factor near unity automatically, no manual switching. Thyristor-switched stages avoid the capacitor inrush that blows fuses on contactor-switched panels; detuned reactors are added where harmonic-heavy loads like VFDs and UPS are already on the bus.

Type
Thyristorised / Contactor / Hybrid

kVAR range
25 to 800 kVAR

Voltage
415 V, 3-phase, 4-wire

Steps
6 / 8 / 12

Standard
IS/IEC 61439-2

Protection
IP54 indoor, IP65 outdoor

Switching
Zero-point (thyristor)

- Thyristor zero-point switching. No inrush current, no capacitor stress on switching
- Detuned reactor options at 7% and 14% for VFD-heavy loads
- Auto changeover between EB and DG source, no manual reset
- Powder-coated CRCA enclosure built for Bengal humidity
- Reference designs CPRI type-tested at 6000A

IS 8623 · IEC 61439-2 · CPRI type-tested · ISO 9001

MCC Panels

Motor Control Centre, IS 8623, 70 kA / 1s withstand

An MCC groups every motor starter on a plant or floor into one panel, each feeder isolated behind its own door so one motor fault or maintenance job doesn't need a full shutdown. Form 4b segregation keeps live busbar separated from the feeder compartment — the detail that actually gets checked in a safety audit.

Standard
IS 8623 / IEC 61439-2

Short-circuit withstand
70 kA for 1 second

Protection
IP54 / IP55

Configurations
Single-front, Double-front

Feeder types
DOL, Star-Delta, VFD, Soft-start

Form segregation
Form 2b / 3b / 4b on request

Integration
BMS and SCADA ready

- Fixed and draw-out feeder options in the same lineup
- Wired for Siemens, Schneider, L&T or ABB switchgear as you spec it
- Vertical bus spacing rated for the fault level, not the datasheet minimum
- Type-tested reference designs available on request

IS 8623 · IEC 61439-2 · CPRI 6000A · ISO 9001

PCC Panels

Power Control Centre, up to 4000 A, IS 8623

A PCC sits upstream of the MCC, taking the incoming supply and splitting it into outgoing feeders for distribution boards, large motors and the MCC itself. At this current rating a fault has to clear in cycles not seconds, so busbar bracing and breaker discrimination are checked against the full prospective short-circuit current, not just the panel's own rating.

Current rating
up to 4000 A

Voltage
415 V, 3-phase, 4-wire

Standard
IS 8623 / IEC 61439-2

Short-circuit withstand
70 kA for 1 second

Protection
IP54 / IP55

Configurations
Single-front, Double-front

- Load scheduling drops non-critical feeders first when EB or DG is overloaded
- CRCA sheet with powder coat, not painted MS. Bengal-climate rated
- Incoming and outgoing feeder layout planned around your operator, not our drawing template
- ACB or MCCB incomer to consultant spec

IS 8623 · IEC 61439-2 · CPRI 6000A · ISO 9001

AMF & DG Sync Panels

AMF plus DG synchronisation, 10 to 2500 kVA

An AMF panel's only job is to start the standby genset and transfer load the moment mains fails, then transfer back and shut down cleanly when mains returns. Sync panels add the extra step of phase-matching multiple DGs, or a DG to the grid, before closing the breaker — get that wrong and you're closing two live sources out of phase onto the same bus.

kVA range
10 to 2500 kVA

Configuration
Double busbar for multi-DG

Controllers
DEIF, ComAp, DSE, Woodward

Modes
AMF, Manual, Load Share, Grid Parallel

Load sharing
Droop or Isochronous, kW plus kVAR

- Multi-DG start, sync and priority handover without an operator on the panel
- EB fail auto-transfer and auto-restore, tuned to your no-break window
- Controller-agnostic. We wire on the brand your service engineer already carries
- FAT before dispatch. Test sheet, SLD, GA and wiring drawings in the file

IS/IEC 60947-6-2 · DEIF / ComAp / DSE certified · ISO 9001

PLC Automation Panels

Multi-brand PLC panels, any major platform

A PLC panel replaces relay logic with a programmable controller, so sequencing changes become a program edit instead of a rewiring job. We wire to the PLC and HMI brand your maintenance team already knows, because a panel nobody on site can reprogram is a liability, not an asset.

PLC brands

Siemens, Allen-Bradley, Mitsubishi, Delta, Schneider

HMI options

Weintek, Siemens KTP, Delta DOP

SCADA

Ignition, WinCC, iFIX ready

Protocols

Modbus TCP/RTU, Profinet, EtherNet/IP, OPC UA

Wiring

IEC ferruled harness, UL 508A on request

- Multi-brand integrator. We spec the right PLC for your plant, not the one we resell
- Baseline of 16DI / 16DO / 8AI, scaling to 128+ point systems
- Redundant CPU and dual power for critical process
- Deliverables in every file: SLD, GA, BOM, wiring, FAT protocol

IEC 61439 · UL 508A on request · ISO 9001

HVAC Control Panels

BMS-ready HVAC starters, VFD provision

HVAC starter panels run chillers, pumps and AHU fans against a building's BMS, typically with Type-2 coordinated starters so a fault lets go clean without damaging the contactor. VFD bypass wiring is built in from day one, so a drive failure doesn't take the whole air-handling unit down with it.

Starter type

Fuse-less, Type-2 coordinated

Integration

BMS via potential-free contacts

VFD

Auto/manual, local/remote, bypass optional

Incomer

MCB, MCCB or MPCB per consultant spec

- Auto and manual variants on the same drawing set
- BMS wiring standard, not an extra line item
- Modular layout so you can add starters on site without a redesign
- Built to Blue Star, DKK and Voltas consultant specs we already know

IEC 61439 · BMS compatible · ISO 9001

AHU & Metering Panels

AHU starter plus tenant metering, BMS-ready

Where an HVAC panel runs the fan, an AHU metering panel accounts for what each tenant or floor actually drew, over Modbus to the building's energy management system. Getting the CT ratio and metering class right the first time avoids a billing dispute six months after handover.

Starter

Fuse-less, Type-2 coordinated

Metering

Multi-function meter per feeder (Schneider, Selec or L&T)

Comms

Modbus RTU / TCP on RS-485 or Ethernet

Integration

BMS with feeder-level energy monitoring

- Meter per feeder for tenant billing or EnMS reporting
- VFD bypass wired in when the drive matters more than the driven load
- Local/remote and auto/manual selectors on the door
- Terminal blocks sized for site retrofits, not just the day-one BOQ

IEC 61439 · BMS and energy metering · ISO 9001

Industrial Control Desks

Floor or wall-mount desks, IP-55, iron alloy

A control desk puts every start/stop, selector and mimic indicator for a process line at one operator position, laid out to the process rather than a standard template. IP-55 rating and an iron-alloy body hold up on a shopfloor where the desk gets leaned on daily.

Protection
IP-55

Mounting
Floor-standing or Wall-mounted

Construction
Iron alloy, powder-coated

Operation
Buttons with optional mimic panel

- Dimensions and cutouts drawn to your operator ergonomics
- Mimic and indicator layout planned before we cut the plate
- Built to move. Sturdy, but liftable on shopfloor trolley

IEC 61439 · IP-55 · ISO 9001

DG Control Panels

Single-DG monitoring and control, domestic to commercial

A single-DG control panel is the simpler sibling to an AMF panel: monitoring and manual or auto-start control for one genset, without the load-transfer logic a full AMF panel needs. Right-sized for a housing complex or small factory where a dedicated changeover panel would be over-spec.

Application
Single-DG monitoring and control

Segments
Domestic, Commercial, Small-industrial

Controllers
DSE 4520, ComAp InteliLite or equivalent

Construction
Powder-coated CRCA, corrosion-resistant

- Compact door layout that opens fully in a tight DG room
- Long service life on the controller and the door hardware, not just the enclosure
- Priced for a small building, not scaled down from an industrial panel

IS/IEC 60947 · DSE / ComAp certified · ISO 9001

Fire Pump Control Panels

Auto plus manual mode, 415 V, industrial-grade

Fire pump and detection control panels on a chemical, pharma or plastics site need to work in auto mode without an operator present, and in manual override when one is. Enclosure material and cable entry detail matter more here than on a standard panel, since the atmosphere it sits in is usually the reason the panel exists.

Voltage
415 V

Modes
Auto plus Manual

Mounting
Wall-mounted or Floor-mounted

Application
Chemical, Plastic, Pharma, Oil Refineries

- Auto activation on pressure drop, with manual override at the door
- Corrosion-resistant enclosure for chemical and coastal sites
- Terminated for the field wiring plan you already have

IS 2189 · NBC 2016 · ISO 9001

Power Distribution Panels

Modular PDBs for public and industrial distribution

A PDB takes one incoming supply and splits it into multiple protected outgoing ways, sized to the load list and segregated so a downstream fault trips only its own way. It's the panel most buyers under-spec — the spare-way count that looked generous at handover rarely survives five years of additions.

Design

Modular, segregated compartments

Custom sizing

Yes, to client single-line diagram

Construction

Rust-resistant powder-coated CRCA

- Compartment spacing that a maintenance sparky can actually work in
- Bus and feeder segregation kept honest, not just labelled
- Rust-resistant body rated for our monsoon, not a marketing brochure

IS 8623 · IEC 61439 · ISO 9001

STP / ETP / WTP Panels

Custom control panels for sewage, effluent and water

Water and effluent plant panels run pumps, blowers and dosing motors against level and pressure interlocks — usually relay logic for a simple lift station, PLC sequencing for a multi-stage plant. IP54 enclosures and corrosion-resistant hardware are standard here, not an upgrade, given where the panel lives.

Applications

STP (Sewage), ETP (Effluent), WTP (Water)

Control

Relay logic or PLC-based sequence

Testing

Full parameter test before dispatch

- Sequence control, pump start-stop, level and pressure interlocks built to your P&ID
- Relay logic for small plants, PLC and HMI upgrade path when the site scales
- Panel wiring rated for STP room humidity, not office weather

IEC 61439 · WBPCB / CPCB aligned · ISO 9001

Custom & Specialised Panels

Burner, boiler, BMB, ash handling. Built to your process

Burner, boiler, ash-handling and material-handling lines rarely match a catalogue panel, so these are built against the customer's own SLD and control philosophy rather than a standard spec sheet. If your process doesn't fit one of the other twelve panel types, this is the one we quote.

Types

Burner, Boiler, BMB, Ash Handling and more

Applications

Furnace, Steam, Small MCC, Cement and Ash

Delivery

Engineered-to-order

- Interlocks and sequencing written for the process, not just the electrical BOQ
- Sensor and actuator terminations planned with your instrumentation vendor
- Documentation set for every job: SLD, GA, wiring, I/O list, FAT protocol

Engineered-to-order · IEC 61439 base · ISO 9001

WHERE OUR PANELS RUN

Industries we build for.

Thirty-four years of panels built to spec for specific sites, not a catalogue. Eight sectors below cover most of what we build; if yours isn't listed, it's probably still a custom panel we can quote.

Cement, Steel & Material Handling

Conveyor, crusher and ash-handling duty: high motor counts, dusty sites, heavy restart current. We build MCC and PCC panels with Form 4b segregation so one feeder fault does not drop the line.

Water, Effluent & Sewage Treatment

STP, ETP and WTP panels for municipal and industrial plants: pump and dosing motor control, filter and aerator sequencing, IP54 enclosures for a wet plant room.

Chemical, Petrochemical & Pharma

Batch process control and fire/gas safety panels for chemical and pharma sites, built to the segregation and labelling a safety audit actually checks for.

Food & Beverage Processing

PLC-driven batching, washdown-duty enclosures and power factor correction for plants running compressors and refrigeration most of the day.

Commercial Real Estate & Hospitality

Malls, hotels and hospitals: HVAC and AHU control, metering panels per tenant or floor, and DG/AMF changeover that switches over before anyone in the lobby notices.

Data Centres & Telecom

Uptime depends on the changeover panel, not just the generator. AMF/DG sync panels with no manual reset step, plus APFC to keep the incoming supply penalty-free.

Utilities, Residential & Small Industry

Single-DG control panels and distribution boards sized for a housing complex, a small factory or a utility substation, not over-specified for either.

Custom Process Plants

Jute and textile mills, boiler houses, furnace and burner lines, oil refineries: non-catalogue panels built off your SLD, not ours.

TRACK RECORD

Panels running on site for.

A selection of the organisations currently running AM Electricals panels on site. Full reference list and site visits available on request.

L&T Construction

Patton

Qualicom Solutions

Steam & Control Systems

Ovaron Infrastructure & MEP

Herringbone Technical Services

Wesman

Also: Ipsen

GET IN TOUCH

Send the spec. We reply in a day.

SLD, load list, or a photo of the existing panel. Fastest is WhatsApp. Everything is reviewed by our founder before we quote.

Phone	+91-7003707013
WhatsApp	Click to chat — wa.me/917003707013
Email	amelectricals@yahoo.co.in
Factory	7S Kamardanga Road, Near Anandapalit, Kolkata 700046, West Bengal
Website	amelectricals.co.in